

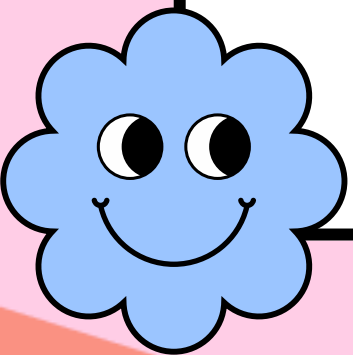
Python Basics + NumPy

Introduction

SESSION 2

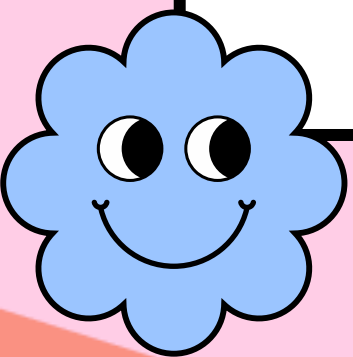
ACCESS 6.0 DATA SCIENCE TRACK

INSTRUCTORS : FAVOUR & MIRIAM



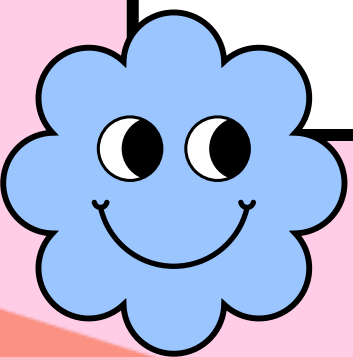
What We'll Cover Today:

- Python fundamentals: the building blocks you'll use every single day
- NumPy: why it's essential for data analysis
- Hands-on practice in Google Colab
- GitHub integration for your portfolio



Why This Matters:

Every data science project starts with understanding how to manipulate data. Python gives you the tools, and NumPy makes those tools incredibly powerful and fast.

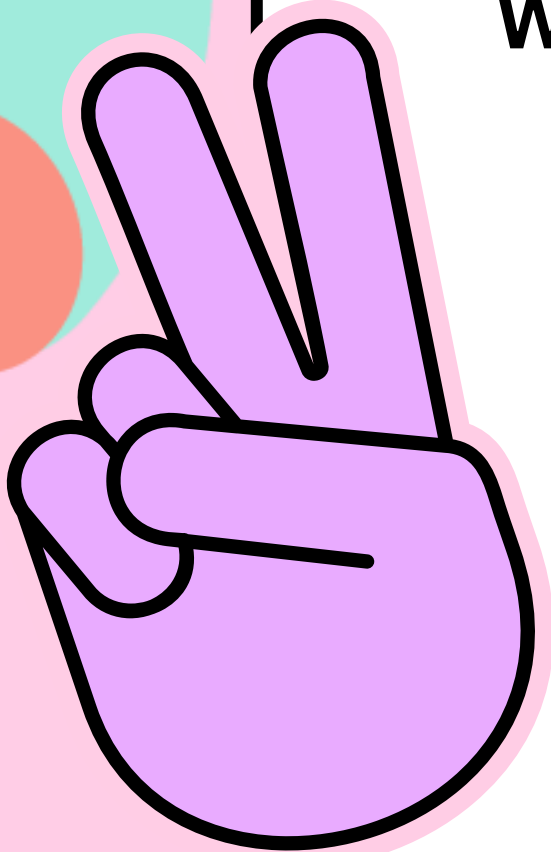


Setting Up Google Colab

Google Colab is a free, cloud-based Jupyter notebook environment that requires no setup. It's perfect for learning and sharing your work.

Why Google Colab:

1. No installation required - runs in your browser
2. Free access to GPUs for intensive computations
3. Easy to share notebooks with others
4. Integrates with Google Drive for storage
5. Pre-installed with NumPy, Pandas, and other essential libraries

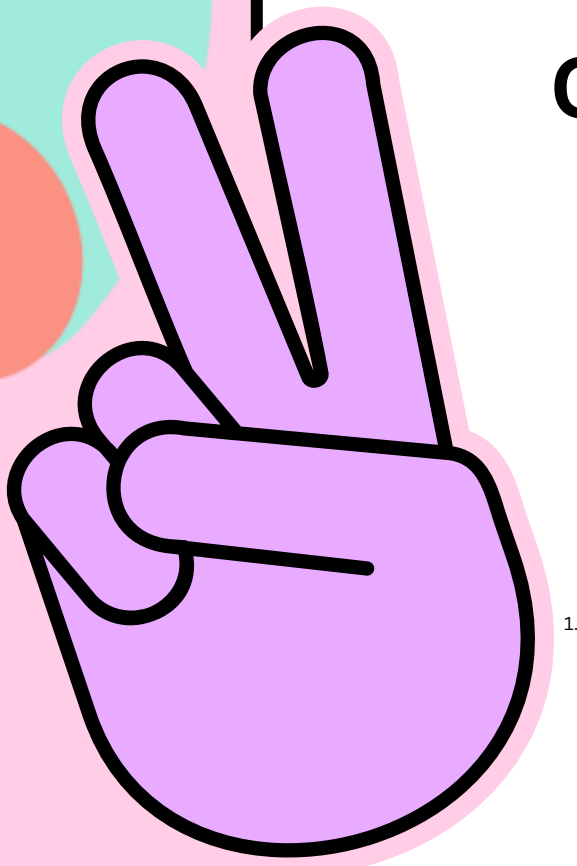


Getting Started:

1. Go to colab.research.google.com
2. Sign in with your Google account
3. Click "New Notebook"
4. You're ready to code!

Colab Cells:

- Code cells: Where you write and run Python code
- Text cells: Where you write explanations using Markdown
- Press Shift+Enter to run a cell
- Click "+ Code" or "+ Text" to add new cells



Steps to Push Colab Notebook to GitHub:

1. In GitHub (one-time setup):

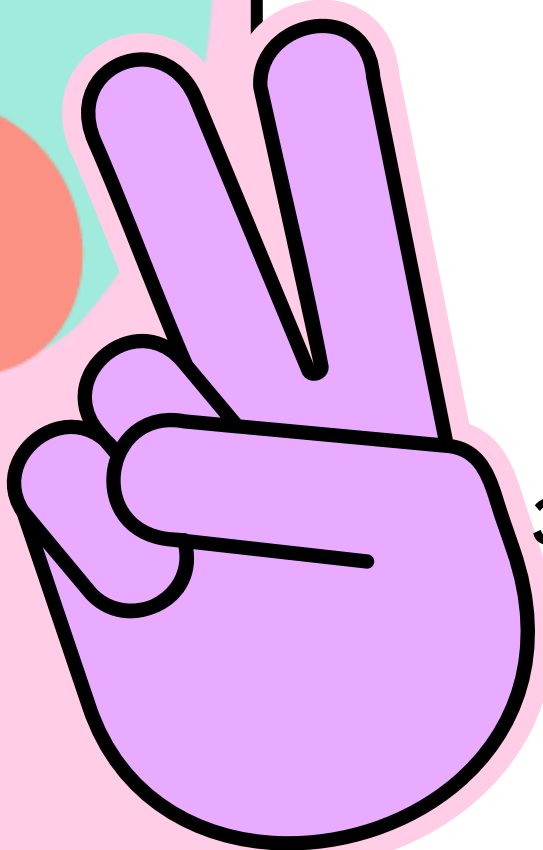
- Go to github.com and create an account if you don't have one
- Click the "+" icon and select "New repository"
- Name it something like "data-science-practice"
- Add a description: "Python and NumPy exercises from data science course"
- Make it Public (so employers can see it)
- Check "Add a README file"
- Click "Create repository"

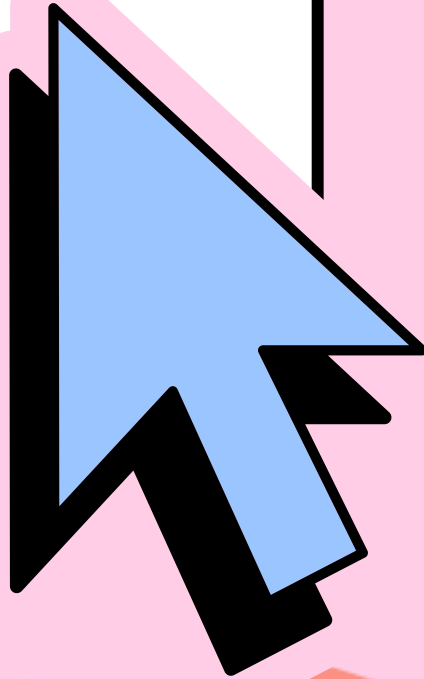
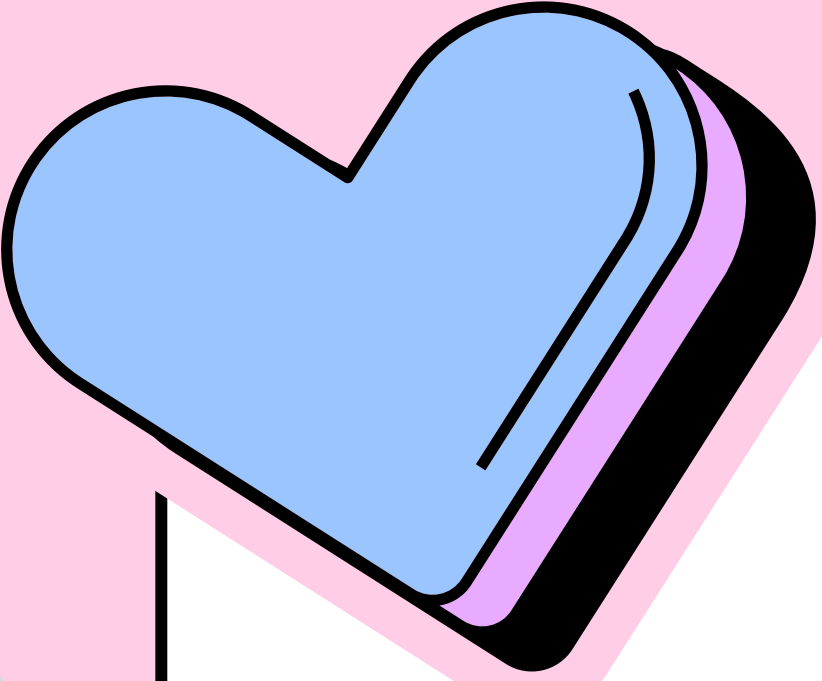
2. In Colab:

- Open your notebook
- Click "File" → "Save a copy in GitHub"
- Authorize Colab to access your GitHub account (first time only)
- Select your repository
- Add a commit message (e.g., "Add Session 2 assignment - transaction analysis")
- Click "OK"

3. Verify:

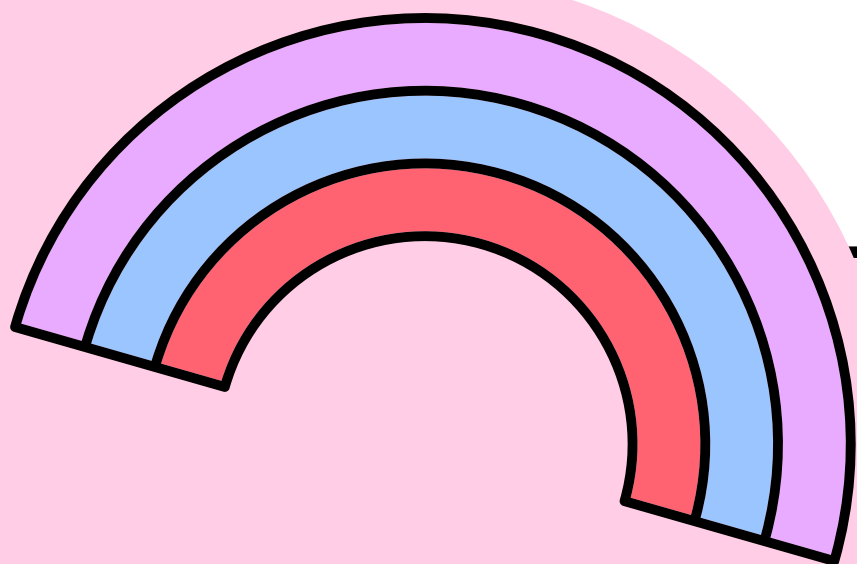
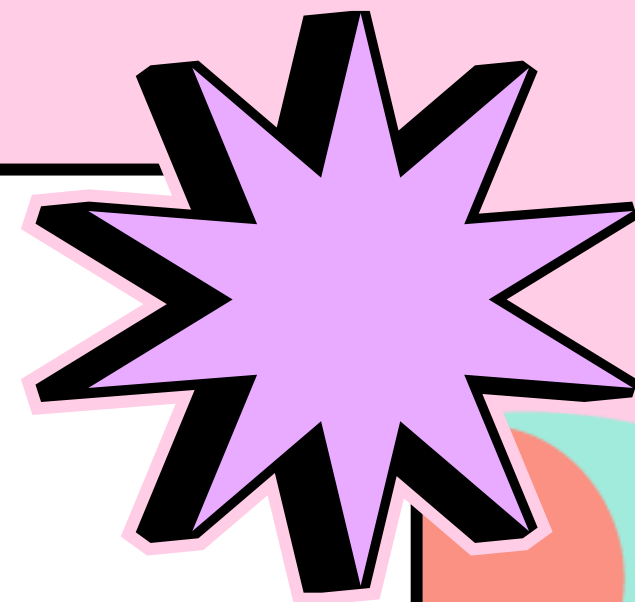
- Go to your GitHub repository
- You should see your notebook file (.ipynb)
- Anyone can now view your code and see your work





Let's
Start!

Any Questions?



ASSIGNMENT

Assignment:

Write a function that takes a list of transaction amounts and prints:

Total sum

Average

Number of transactions

Save and push the notebook to your GitHub repo.

